

2025 PSYC489 Research Project

Into the Wormhole: Exploring Ketamine's Neural Mechanisms in Planarian Models of
Substance Use

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Abstract

Substance use disorders (SUDs) are chronic, relapsing conditions with high relapse rates, underscoring the need for therapies targeting core neurobiological mechanisms. This study is the first to investigate whether ketamine attenuates methamphetamine-induced conditioned place preference (CPP) and whether ketamine independently induces CPP in planaria. In Experiment 1, planaria exposed to ketamine (10 μ M or 30 μ M) did not produce detectable CPP, indicating minimal intrinsic rewarding effects at these concentrations. Experiment 2, which assessed post-conditioning ketamine treatment (10 μ M or 30 μ M) following methamphetamine CPP (10 μ M), showed a near-significant attenuation of CPP ($p = .053$), whereas 30 μ M had no effect. Methamphetamine reliably induced CPP, confirming conserved dopaminergic reinforcement in planaria. These findings suggest that ketamine's anti-addictive effects may require neural mechanisms absent in this model. Collectively, these findings confirm that planaria reliably model dopaminergic reinforcement but are limited for assessing cortical-glutamatergic modulation, delineating translational boundaries of NMDA antagonist effects across species.

Introduction

Substance Use Disorder and Current Treatment Approaches

Substance use disorders (SUDs) are complex, chronic conditions impacting more than 64 million people internationally (UNODC, 2023, p.58). SUDs result in higher rates of morbidity and mortality, homelessness and poverty (Baptiste-Roberts & Hossain, 2018), social exclusion (Wesselmann & Parris, 2021), family disruption (Girling et al., 2006), and significant economic burden on healthcare and criminal justice systems (Fardone et al., 2023). Despite these far-reaching impacts, only about 9% of individuals with SUDs access treatment, and relapse rates remain high – 40–80% within the first year of standard treatment (McLellan et al., 2000). These sobering statistics highlight the need for effective, accessible therapies that target the core neurobiology of addiction.

Current behavioural and pharmacological treatments often show limited long-term success, primarily due to challenges related to adherence, accessibility, and relapse prevention (Krotter et al., 2024; Magill et al., 2023; Ray et al., 2018). Inadequate psychosocial support and vocational rehabilitation also contribute to poor outcomes, despite their importance for sustained recovery (Azmi et al., 2018). Additionally, most pharmacotherapies target single receptor systems – such as opioid, nicotinic, or GABA – leaving the shared neurobiological mechanisms underlying compulsive drug use largely unaddressed. Emerging agents targeting glutamatergic pathways, such as ketamine, hold particular promise for clinicians and researchers (Jones et al., 2018; Kryst et al., 2020).

Ketamine as a Novel Therapeutic Agent for SUDs

Ketamine is an *N-methyl-D-aspartate* (NMDA) antagonist. Originally developed as an anaesthetic in the 1960s, it is now being explored as a novel option for addressing addiction (Hirota & Lambert, 2022). While its dissociative properties have required regulatory oversight, recent research demonstrates its rapid and sustained antidepressant effects (Li & Vlisides, 2016) and its ability to reduce craving and relapse in various SUDs (Dakwar et al., 2014; Ezquerra-Romano et al., 2018; Kelson et al., 2023). These characteristics warrant further attention from researchers and treatment providers.

High rates of co-occurrence between SUD and major depressive disorder (MDD) intensify both symptom severity and the risk of relapse (Blanco et al., 2012; Danovitch et al., 2017; Hunt et al., 2020). Factors contributing to this interplay include self-medication attempts (Khantzian, 1997; Turner et al., 2018), reciprocal causation, and shared biological vulnerabilities (Blanco et al., 2012). This association supports a rationale for considering ketamine's role in treating the intertwined neurobiology of addiction and depression, an approach clinicians may find relevant as it leverages enhanced neuroplasticity to support complex cases (Janssen-Aguilar et al., 2025).

Ketamine's Mechanism of Action

Ketamine blocks NMDA receptors on prefrontal GABAergic interneurons, disinhibiting pyramidal neurons and triggering a glutamate surge that increases the activity and number of *α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid* (AMPA) receptors (Gerhard et al., 2020; Pothula et al., 2021). This cascade upregulates brain-derived neurotrophic factor (BDNF) translation and activates the mammalian target of rapamycin (mTOR) pathway (Duman et al., 2019; Li et al., 2010; Zhou et al., 2014), thereby promoting synaptogenesis and repairing disruptions caused by chronic

drug exposure (Ezquerro-Romano et al., 2018). Ketamine also modulates dopamine (Kokkinou et al., 2017), serotonin (Nepton et al., 2024), and neuroinflammatory systems (Jóźwiak-Bębenista et al., 2024), all of which are critical for mood and reward processing.

Ketamine exhibits an inverted U-shaped dose-response curve (Choudhury et al., 2021; Vestring et al., 2024). Low, subanaesthetic doses induce neuroplastic changes with mild perceptual effects (Chowdhury et al., 2017). In contrast, high doses can produce pronounced dissociation, cognitive impairments, and physical side effects such as cardiovascular disturbances and neurotoxicity (Green & Coté, 2009; Li et al., 2025). Precise dosing within the therapeutic window maximises antidepressant and anti-addictive benefits while minimising risks.

Ketamine's acute effects resolve rapidly due to its short half-life (~2-4 hours), while antidepressant benefits can last up to a week (Murrough et al., 2013; Zarate et al., 2006). Importantly, the neuroplastic mechanisms underlying ketamine's antidepressant effects are also thought to enable the disruption of maladaptive drug-cue associations formed through repeated drug exposure (Ezquerro-Romano et al., 2018). Through these glutamatergic pathways, ketamine can selectively interfere with the reconsolidation and extinction of drug-related memories, targeting the core mechanisms of addiction without inducing general amnesia (Das et al., 2019). These time-sensitive neuroplastic changes may 'rewire' pathological neural circuits, allowing ketamine to reduce craving and relapse risk, highlighting its potential relevance in SUD treatment.

However, ketamine has addictive liability in humans. Careful experimental controls are necessary to separate anti-addiction effects from intrinsic reward potential. In mammals, ketamine induces conditioned place preference (CPP), a behavioural

model for assessing drug reward and addiction liability (Chang et al., 2019; Peng et al., 2023; Suzuki et al., 1999), paralleling clinical evidence of dependence, craving, and relapse (Marongiu et al., 2025). However, the findings of abuse potential in mammalian models are often inconsistent, highlighting difficulties in translating across species.

Cross-Species Variation in NMDA Pharmacology

While ketamine's mechanism of action as an NMDA receptor antagonist is well established in mammals, cross-species differences significantly influence pharmacological and behavioural outcomes (D'Aniello et al., 2007; Eugenin et al., 2011; Lei & Tejani-Butt, 2010). NMDA receptor subunits and extracellular conditions, such as magnesium concentrations, vary across species, including invertebrates like planaria. These variations highlight the absence of complex neural circuitry, such as prefrontal cortical structures, in simpler organisms, such as planaria (Pagán, 2017). This lack can lead to the absence of ketamine's full range of effects as seen in mammals.

These molecular differences may explain why ketamine induces an inverted U-shaped dose-response curve observed in mammals, where low doses promote neuroplasticity and reinforcing effects and higher doses cause sedation or neurotoxicity (Chowdhury et al., 2017; Green & Coté, 2009; Li et al., 2025; Zhou et al., 2014). These changes may fail to produce analogous behavioural responses in organisms lacking complex neural circuitry. Recognising these factors is critical for interpreting ketamine's behavioural and therapeutic profiles; it provides a foundation for comparing it with other NMDA antagonists in preclinical models, helping to manage expectations about the translation from animal studies to human clinical applications.

Ketamine Compared to Other NMDA Antagonists Across Animal Models

Ketamine belongs to a broader class of NMDA receptor antagonists that vary in affinity, selectivity, kinetics, and clinical relevance. While several agents can modulate neuroplasticity, ketamine distinguishes itself through efficiency and better tolerability.

MK-801 produces prolonged NMDA blockade linked to cognitive deficits and neurotoxicity, limiting its translational applicability compared to ketamine's more controlled inhibition (Kovacic & Somanathan, 2010). MK-801 also demonstrates reinforcing properties independent of dopaminergic signal modulation, evidenced by rats' self-administering injections into the nucleus accumbens and frontal cortex even when given dopamine antagonists (Carlezon & Wise, 1996). Clinically, ketamine's superior efficacy and tolerability make it a promising candidate for therapeutic development.

Despite limited evidence, NMDA antagonists have demonstrated anti-addictive potential. Noncompetitive antagonists – including memantine and ketamine – reduce alcohol consumption in alcohol-preferring rats, though memantine's effect appears alcohol-specific (Sabino et al., 2013). Ketamine's mTOR-mediated mechanism supports broader therapeutic potential across SUDs, including blocking morphine-induced CPP in mice (McKendrick et al., 2020) and preventing stress-induced opioid reinstatement (Michael et al., 2025). Collectively, these findings underscore ketamine's ability to induce controlled neuroplasticity while maintaining clinical tolerability, supporting its promise as a treatment for SUDs.

Although extensive research has been conducted with mammals, ketamine's dual rewarding and anti-addictive effects remain unexplored in high-throughput invertebrate models such as planaria. With their conserved neurotransmitter systems,

planaria offer an ethically responsible and efficient preclinical model, providing valuable insight into ketamine's impact on drug reward and associative learning at the most fundamental level.

Planaria as a Model Organism in Addiction Research

Planaria flatworms serve as an effective model for addiction-related behaviours. Although their nervous system is simple, planaria express functional serotonin (Fernandes et al., 2003), glutamate (Rawls et al., 2006), dopamine (Venturini et al., 1989), and acetylcholine receptors (Buttarelli et al., 2000). They also exhibit behavioural responses to drugs of abuse analogous to those seen in mammals (Raffa et al., 2013; Rawls et al., 2007). Notably, planaria have functional NMDA receptors, as evidenced by NMDA-induced seizure-like activity that is attenuated by MK-801 (Rawls et al., 2009).

The NMDA antagonist LY 235959 also reduces abstinence-induced withdrawal symptoms in planaria following cannabinoid exposure, indicating that NMDA receptor activation contributes to dependence and withdrawal behaviours (Rawls et al., 2007). Planaria also demonstrate CPP for methamphetamine and other stimulants, withdrawal-like behaviours following cessation, and sensitisation with repeated exposure (Hutchinson et al., 2015; Raffa et al., 2013; Ramirez, 2025).

However, despite conserved neurotransmitter systems, planaria lack higher-order brain structures, such as the prefrontal cortex (Pagán, 2017), which constrains their ability to model the complex motivational and affective components of addiction. While valuable for probing associative learning and dopaminergic reinforcement mechanisms, planaria models cannot fully capture the cognitive and emotional landscape of SUDs (Hutchinson et al., 2015). This distinction helps to prevent over-

interpretation of findings by focusing on the neurochemical underpinnings rather than assuming direct parallels with human addiction experiences.

Given the suitability as a preclinical model, the present study examines methamphetamine, which is the most widely used illicit stimulant in Aotearoa New Zealand after marijuana (Yasbek et al., 2022). Methamphetamine's well-characterised dopaminergic mechanism and its consistent induction of CPP in planaria at 10 μ M (Ramirez, 2025; Sacavage et al., 2008) support its selection as an appropriate agent for modelling addictive behaviours and assessing the modulatory effects of ketamine.

Methamphetamine Use in the Context of Aotearoa New Zealand

Recent wastewater analysis by the New Zealand Police (2025) reported a 96% increase in methamphetamine use between 2023 and 2024, reflecting a two-decade trend of escalating addiction, mental health crises, and social disruption (National Drug Intelligence Bureau, 2023; Yasbek et al., 2022). Methamphetamine's widespread impact, including disproportionate harm within Māori communities due to systemic inequities and historical trauma, underscores its significance as both a public health and socio-cultural concern (Yasbek et al., 2022). These factors, along with its consistent behavioural effects in planaria, support the selection of methamphetamine as the focal drug for this study.

Conditioned Place Preference (CPP) as an Experimental Paradigm

CPP is a well-validated paradigm for assessing the rewarding or aversive properties of drugs. Organisms are conditioned to associate specific environmental cues with drug exposure, and preference for that context is used as a measure of the drug's motivational *wanting* effect (McKendrick & Graziane, 2020).

CPP has been repeatedly validated in planaria (Adams & Byrne, 2019; Hutchinson et al., 2015; Raffa et al., 2013). This makes it a suitable model for assessing methamphetamine-induced reward and for determining whether ketamine can mitigate this preference while controlling for its inherent rewarding properties.

Careful consideration of planaria as a preclinical model, the well-characterised effects of methamphetamine, and the validated CPP paradigm provides a strong foundation for investigating ketamine's dual reward and anti-addictive properties. Despite promising evidence from mammalian studies, questions remain regarding ketamine's safety, efficacy, and dose-dependent effects in simpler organisms, highlighting the need for systematic evaluation in a high-throughput model.

Research Aims and Hypotheses

To our knowledge, this study represents the first systematic investigation of ketamine's rewarding and anti-addictive effects in planaria. Experiment 1 tests whether ketamine induces CPP in planaria, thereby indicating it has intrinsic rewarding effects in this model. The central hypothesis posits that planaria exposed to ketamine will spend more time in the ketamine-paired environment than controls, with the magnitude of this preference expected to show dose-dependence. This dose-related relationship is hypothesised to exhibit an inverted U-shaped pattern, wherein low to moderate concentrations of ketamine produce the highest preference, as established in mammalian models. This prediction is based on preclinical studies in rodents demonstrating that ketamine produces a measurable rewarding response at low doses via transient dopaminergic activation. In contrast, higher doses are less effective or aversive (Choudhury et al., 2021). Thus, Experiment 1 is designed to test the existence of dose-dependent ketamine-induced CPP in this invertebrate model.

Experiment 2 examines whether post-conditioning ketamine treatment reduces preference for a methamphetamine-paired context. It is hypothesised that ketamine will attenuate methamphetamine-induced CPP in a dose-dependent manner, consistent with its U-shaped efficacy profile. This outcome aligns with existing research indicating that ketamine's modulation of glutamatergic and BDNF-mTOR signalling pathways can interfere with drug-cue memory reconsolidation, thereby reducing relapse rates in rodent models and humans (Das et al., 2019; Ezquerra-Romano et al., 2018).

Method

Design

Two experiments were conducted using a CPP paradigm to examine ketamine's effects on reward-related behaviour in planaria. Experiment 1 tested ketamine alone for CPP induction; the design was a 2 (time: pre- vs. post-conditioning; within-subjects) $\times$ 3 (treatment: control, 10 μ M ketamine, 30 μ M ketamine; between-subjects) mixed-design, with time spent on the drug-paired surface as the primary dependent variable.

Experiment 2 evaluated ketamine's ability to attenuate methamphetamine-induced CPP. The design was a 3 (time: pre-, post-conditioning, post-treatment; within-subjects) $\times$ 3 (treatment: control, 10 μ M ketamine, 30 μ M ketamine; between-subjects) mixed-design, with the same dependent variable.

Subjects and Housing

In each experiment, thirty-six planaria ($N = 36$) were randomly assigned to three groups using Research Randomizer (Urbaniak & Plous, 2013): 10 μ M ketamine ($n = 12$), 30 μ M ketamine ($n = 12$) or control (vehicle only; $n = 12$). In Experiment 1, one planaria (#5) from the 30 μ M group was lost on Day 8 and excluded from all subsequent analyses, resulting in $n = 11$ for this group.

A sample size of 12 planaria per group was chosen based on prior planaria CPP studies demonstrating reliable CPP detection with similar group sizes (Hutchinson et al., 2015; Ramirez, 2025). Formal power analyses were not performed because relevant data on ketamine's effects in planaria are currently unavailable; however, these group sizes balance logistical feasibility and statistical sensitivity for medium-to-large effects.

All planaria were housed individually in 12-well plates containing 3 mL of planaria water (dechlorinated tap water; pH ~7.6), replaced daily. Subjects were acclimated for 24 hours prior to testing and maintained in a darkened environment at room temperature (22 ± 2 °C). Planaria were starved for 24 hours prior to testing in Experiment 1, and 48 hours in Experiment 2; this time variation was due to differences in experimental scheduling. Planaria were maintained under fasting conditions throughout testing to minimise locomotor variability. Individuals of similar size were selected to control for baseline activity and health status.

Solutions

All reagents were obtained from BDG Synthesis (Wellington, New Zealand). All solutions were prepared in distilled water and mixed thoroughly before use. Working concentrations of methamphetamine (10 µM) and ketamine (10 µM, 30 µM) were freshly prepared for each session from respective stock solutions (see Appendix, Table A1). Solutions were replaced between trials to prevent degradation or cross-contamination.

Ketamine concentrations were selected with reference to mammalian studies in which methamphetamine-induced CPP (2 mg/kg) was attenuated at doses of 15-30 mg/kg (Peng et al., 2023). Given planaria's simpler physiology and potential sensitivity

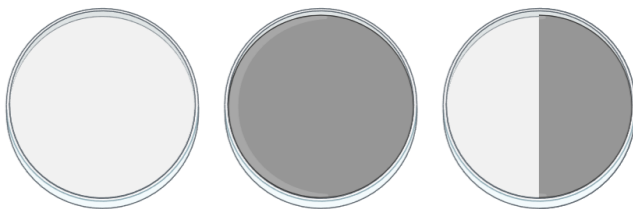
compared to mammals, ketamine concentrations were conservatively reduced to minimise toxicity while maintaining exploratory value.

Apparatus

CPP testing was conducted using 3D-printed circular dishes (6 cm diameter, 0.5 cm wall height) configured as completely smooth, completely rough, or dual-textured (half smooth/half rough) (See Figure 1). The rough surface consisted of uniform, 0.2 mm raised bumps spaced 0.2 mm apart, providing consistent tactile cues.

Experiments were conducted in a dark room under dim red light to minimise phototactic effects. Behaviour was recorded overhead using an Apple iPhone XS Max camera, and videos were subsequently analysed manually to determine time spent on each surface.

Figure 1



Note. Schematic of the conditioned place preference (CPP) apparatus. The white compartment (left) represents the completely smooth context, the grey compartment (centre) represents the completely rough context, and the rightmost arena shows the combined test arena with both compartments present for place preference assessment.

Handling

Planaria were transferred using a pipette, with the liquid fully expelled before placement. This method minimised stress, prevented mechanical injury, and reduced solution dilution.

Procedure

Both experiments employed a CPP paradigm to assess reward- and aversion-related learning in planaria. Experiment 1 tested whether ketamine alone produced CPP, while Experiment 2 examined whether ketamine attenuated methamphetamine-induced CPP. The CPP protocol used in this study was adapted from Mohammed Jawad et al. (2017), which details a robust CPP assay methodology for planaria. The procedure followed established conditioning-testing phases, alternating between distinct drug and control textures to assess preference learning.

Experiment 1: Ketamine CPP

Baseline Phase. On day 1, each planaria underwent a single 10-minute session in a dual-textured dish to establish initial surface preference. Four planaria were assessed simultaneously in 6 mL of planaria water, with nine rounds conducted ($N = 36$). Manual scoring determined each subject's least preferred texture (smooth or rough), which was designated as the ketamine-paired context for subsequent conditioning.

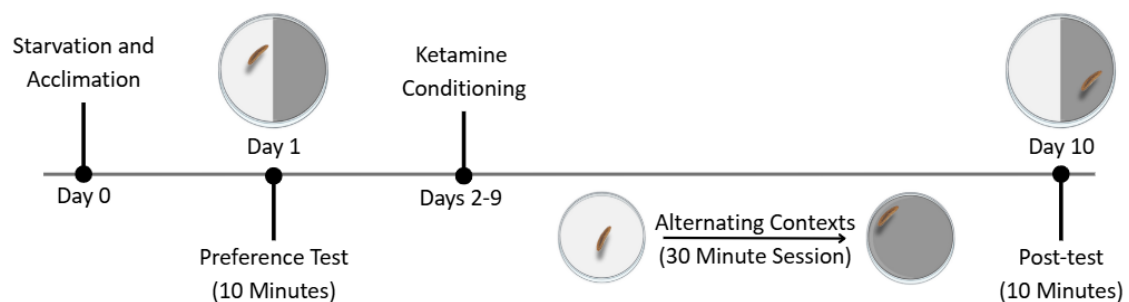
Conditioning Phase. Conditioning occurred over eight consecutive days (Days 2-9), with exposure alternating between the ketamine-paired and vehicle-paired textures. Exposure order and surface pairing were fully counterbalanced across individuals to control for order effects and innate texture preferences. The control group received vehicle-only exposure on both surfaces. Sessions lasted 30 minutes using

freshly prepared solutions. Between sessions, planaria underwent a 10-minute washout in planaria water to mitigate drug carryover. Dishes were rinsed thoroughly between sessions.

Post-Test. On Day 10, a 10-minute drug-free preference test in the dual-textured dish was conducted 24 hours after the final conditioning session. The primary dependent variable was the change in time spent on the ketamine-paired texture pre- versus post-conditioning.

Figure 2

Experimental Timeline for Experiment 1: Ketamine Conditioned Place Preference in Planaria



Note. Timeline for ketamine CPP in planaria. Day 0 represents 24 hours of starvation and acclimation before the baseline preference test (Day 1). Days 2-9: Planaria alternately exposed to ketamine- or vehicle-paired contexts for 30-minute sessions, with a 10-minute washout after each session. Day 10: Drug-free preference test.

Experiment 2: Methamphetamine CPP + Ketamine Treatment

Baseline Phase. Baseline preference was assessed on Day 1 in a 10-minute session in a dual-textured dish. A methodological adjustment reduced the solution volume from 6 mL to 5 mL to improve tactile feedback. Four planaria were tested

simultaneously across nine rounds ($N = 36$). The least preferred texture was designated as the methamphetamine-paired context.

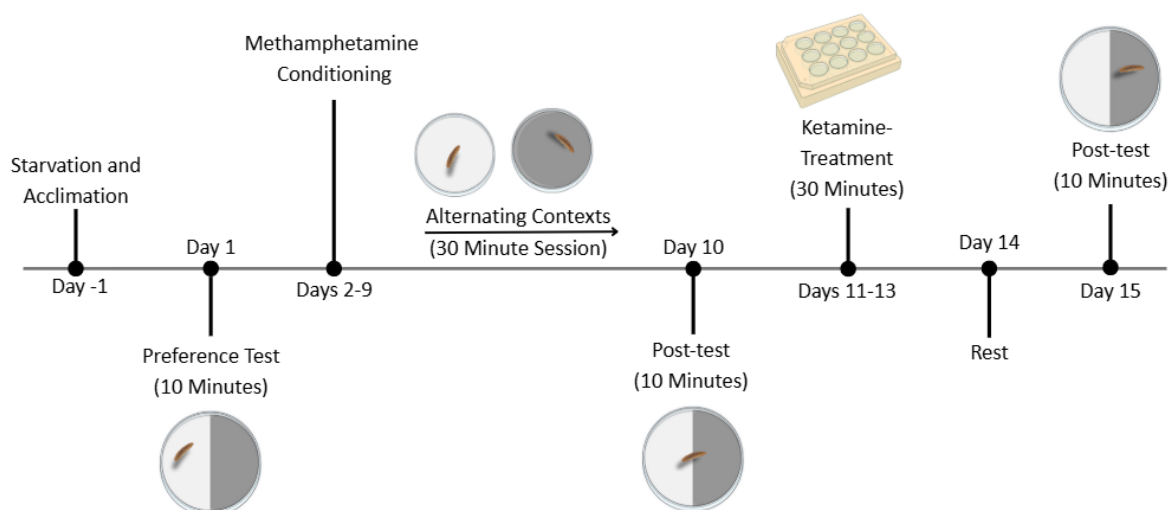
Conditioning Phase. Planaria underwent eight consecutive days (Days 2-9) of alternating 30-minute exposures to methamphetamine-paired and vehicle-paired surfaces. Order was counterbalanced, with nineteen experimental planaria (10 μ M: $n = 10$; 30 μ M: $n = 9$) starting in the methamphetamine-paired context. The control group received vehicle-only exposure. Between sessions, subjects underwent a 10-minute washout, and dishes were rinsed to prevent cross-contamination.

Post-Conditioning Test. Twenty-four hours after the final methamphetamine conditioning session (Day 10), animals underwent a drug-free preference test in the dual-textured dish (5 mL planaria water) to confirm the establishment of CPP. The primary outcome was the change in time spent on the methamphetamine-paired texture compared to baseline.

Ketamine Treatment. On Days 11-13, experimental groups were exposed to either 10 μ M or 30 μ M ketamine to assess attenuation of methamphetamine-induced CPP. Controls received the vehicle only. Sessions were conducted for 30 minutes each day using 12-well dishes covered with orange masking tape to differentiate them from their housing compartments. All planaria ($N = 36$) were treated simultaneously.

Rest Period. A 24-hour rest period (Day 14) was implemented to limit drug carryover effects.

Post-Treatment Preference Test. On Day 15, a final 10-minute drug-free preference test assessed the change in time spent on the methamphetamine-paired context post-ketamine treatment.

Figure 3*Experimental Timeline for Experiment 2: Methamphetamine Conditioned Place**Preference and Ketamine-Treatment Preference in Planaria*

Note. Timeline for ketamine CPP in planaria. Day -1 represents 48 hours of starvation and acclimation before the baseline preference test (Day 1). Days 2-9: Planaria alternately exposed to methamphetamine- or vehicle-paired contexts for 30-minute sessions, with a 10-minute washout after each session. Day 10: Drug-free preference test. Days 11-12: Ketamine or vehicle treatment (administered in 12-well plates covered in orange masking tape). Day 14: Rest days. Day 15: Final drug-free preference test.

Data Analysis

Behavioural data were scored manually by the researcher. Analysis using EthoVision XT15 software (Noldus Information Technology, Wageningen, the Netherlands) was initially planned but could not be reliably performed as the video recordings lacked sufficient lighting contrast. Recorded videos were reviewed individually on an Apple iPhone XS Max. Due to planaria's slow locomotion, videos were sped up during periods of inactivity to optimise scoring efficiency. The researcher

recorded the location of each planaria relative to the compartment boundaries and measured the time (seconds) spent in each compartment by noting when a specimen crossed the apparatus's central dividing line. These times were then cumulatively tallied to calculate the primary dependent variable: total time spent on the drug-paired surface during testing sessions. This manual scoring approach enables detailed, accurate measurement despite the absence of automated tracking software.

Experiment 1 data were analysed using a 2 (time: pre/post) $\times$ 3 (group: control, 10 μ M, 30 μ M) repeated-measures ANOVA. Experiment 2 data were analysed using a 3 (time: pre-conditioning, post-conditioning, post-treatment) $\times$ 3 (group: control, 10 μ M, 30 μ M) repeated-measures ANOVA with Bonferroni-corrected post hoc comparisons.

All analyses were conducted in IBM SPSS Statistics for Windows, Version 30.0 (IBM Corp., Armonk, New York), with assumptions of normality and sphericity confirmed prior to analysis.

Effect sizes were calculated manually using established formulas for Cohen's d and partial eta squared, derived from group means and standard deviations where appropriate. While the chosen sample size ($n = 12$) aligns with established protocols in planaria CPP studies, it reduces the statistical power available to reliably detect small-to-moderate effects ($\eta^2 < .06$; $d < 0.50$). This limited power increases the probability of Type II error, necessitating caution when interpreting non-significant results. Consequently, the inability to detect statistically significant effects in some comparisons may reflect insufficient sample size rather than the absence of an actual effect. Replication with larger sample sizes is therefore essential to clarify these findings and to enhance the robustness and generalisability of observed effects.

Results

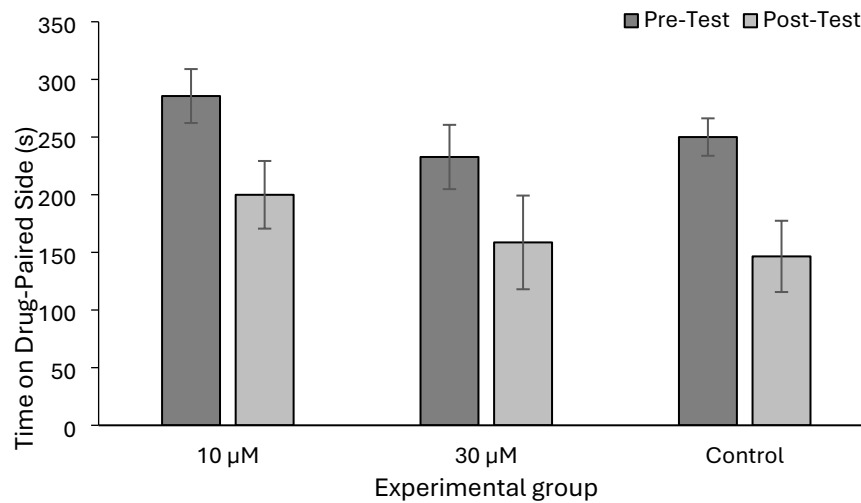
Experiment 1: Ketamine CPP

Planaria in all groups, including those exposed to ketamine, demonstrated a significant reduction in time spent on the drug-paired surface from pre- to post-conditioning (main effect of time: $F(1,32) = 18.24, p < .001, \eta^2 = .186$). However, neither the main effect of group ($F(2,32) = 0.562, p = .576, \eta^2 = .016$) nor the interaction between time and group ($F(2,32) = 0.225, p = .800, \eta^2 = .005$) reached statistical significance, and both effect sizes were small.

Although planaria in both ketamine groups showed a significant reduction in time on the ketamine-paired surface (10 μM : $M \text{ change} = 85.66 \text{ s}$; 30 μM : $M \text{ change} = 74.09 \text{ s}$), a comparable reduction was also observed in the control group ($M \text{ change} = 103.50 \text{ s}$). This uniform decrease across all experimental conditions indicates the reduction in preference was likely driven by procedural or contextual factors inherent to the experimental design rather than by specific effects of ketamine exposure. Thus, there is no evidence for either a dose-dependent or an aversive effect of ketamine under these protocols. Descriptive statistics for all groups at each phase are presented in Table 1, and Figure 4 depicts the observed changes in contextual preference pre- and post-ketamine CPP.

Figure 4

Mean Time (seconds) Planaria Spent in the Ketamine-Paired Compartment Before and After Conditioning



Note. Mean $\pm$ SEM time spent in the drug-paired context before and after ketamine conditioning. Sample sizes were $n = 12$, except the 30 μM ketamine group, which had $n = 11$ due to the loss of planaria #5 on Day 8. Only data from the preferred context were included.

Table 1

Mean Time (Seconds) Planaria Spent on Ketamine-Paired Context Pre- and Post-Conditioning

Experimental Group	Pre-Conditioning			Post-Conditioning	
	<i>n</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
10 μM	12	285.58	91.18	199.92	108.09
30 μM	11	232.72	100.89	158.63	136.46
Control	12	250.00	58.74	146.50	111.80

Note. *M* = mean (seconds); *SD* = standard deviation.

Experiment 2: Methamphetamine CPP + Ketamine Treatment

The repeated-measures ANOVA revealed a significant main effect of time, $F(2,66) = 11.20, p < .001, \eta^2 = .157$, demonstrating that the duration planaria spent on the methamphetamine-paired context changed meaningfully across the three assessment points (pre-conditioning, post-conditioning, and post-ketamine treatment). Specifically, the average time in the methamphetamine context increased substantially following conditioning, confirming the successful induction of CPP. Furthermore, a significant time-by-group interaction, $F(4,66) = 2.80, p = .033, \eta^2 = .078$, demonstrated that the pattern of change over time was not uniform across experimental groups and varied as a function of ketamine treatment assignment. Importantly, the main effect of group was not significant, $F(2,33) = 2.28, p = .118, \eta^2 = .037$, indicating that, prior to conditioning, baseline mean durations in the methamphetamine-paired contexts were statistically equivalent across all three groups. Full descriptive statistics are presented in Table 2, showing that the largest increases from baseline to post-conditioning occurred in the ketamine-treated groups, while Figure 5 visually represents the change in preference across dosing conditions.

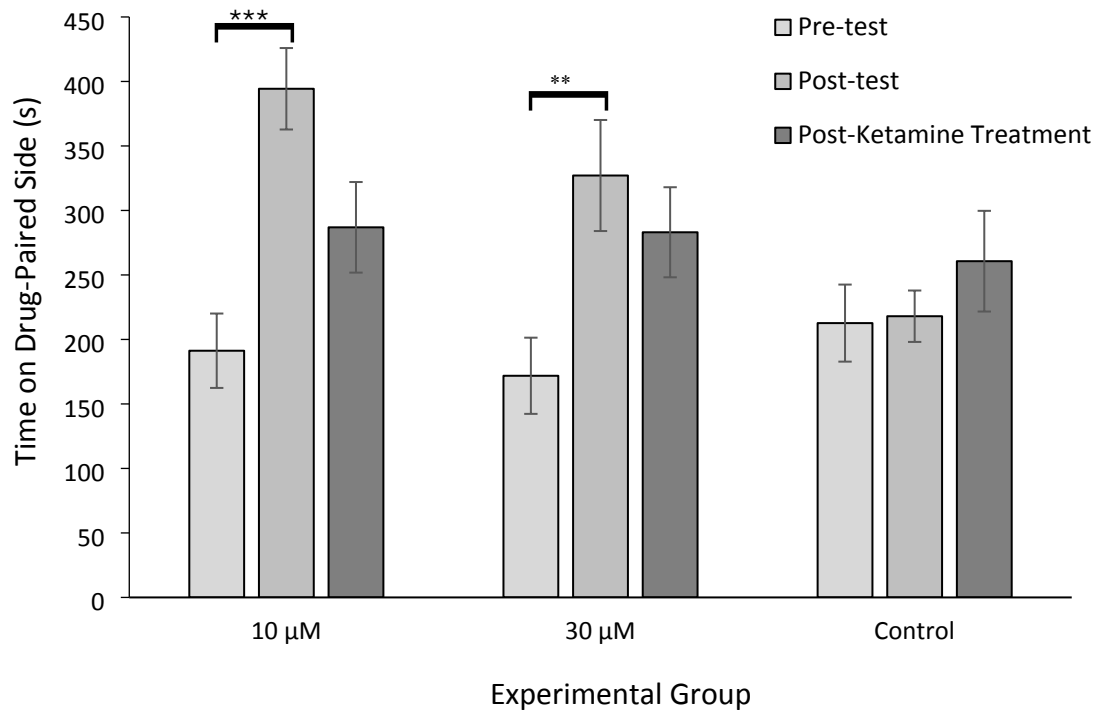
Post hoc analyses with Bonferroni correction revealed that both ketamine treatment groups displayed significant increases from pre- to post-methamphetamine conditioning. Specifically, the 10 μM increased by $M = 203.08$ seconds (95% CI [79.78, 326.38]), $p < .001, d = 1.94$, and the 30 μM increased by $M = 155.25$ seconds (95% CI [16.57, 278.55]), $p = .010, d = 1.21$. In contrast, the control group showed no change ($M = 5.33$ seconds, 95% CI [-128.63, 117.97], $p = 1.00, d = 0.57$).

Following ketamine treatment, post-treatment preference tests indicated a non-significant trend toward attenuation in the 10 μM group ($M = 107.41$ seconds, 95% CI [-1.10, 215.94], $p = .053$) with a large effect size ($d = 0.92$). In contrast, the 30 μM

group exhibited no significant change ($M = 44.00$ seconds, 95% $CI [-64.52, 152.52]$, $p = .942$), though the effect size was small ($d = 0.32$). The observed non-significant reduction within the 10 μM group indicates a potentially significant behavioural effect; however, the relatively small sample size reduced statistical power and heightened the likelihood of a Type II error.

Figure 5

Mean Time (seconds) Planaria Spent in the Methamphetamine-Paired Context Pre-Conditioning, Post-Conditioning, and Post-Ketamine Treatment



Note. Mean $\pm$ SEM time spent on the drug-paired surface showing attenuation trend in the 10 μM group. Sample sizes were $n = 12$. Only data from the preferred context were included. $** = p \leq .01$; $*** = p \leq .001$.

Table 2

Mean Time (Seconds) Planaria Spent on Methamphetamine-Paired Context Pre-Conditioning, Post-Conditioning, and Post-Ketamine Treatment

Experimental Group	Pre-Conditioning			Post-Conditioning		Post-Ketamine	
	<i>n</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
10 μ M	12	191.25	99.90	394.33	109.37	286.92	122.66
30 μ M	12	172.83	102.31	327.08	149.13	283.08	121.95
Control	12	212.67	103.47	218.00	69.03	266.67	135.21

Note. *M* = mean (seconds); *SD* = standard deviation.

Discussion

This study addressed a significant gap in preclinical addiction research by systematically investigating the effects of ketamine on CPP in planaria. The results demonstrated that ketamine alone did not induce CPP in planaria at either 10 μ M or 30 μ M, indicating an absence of intrinsic rewarding effects within this model. The lack of conditioned reinforcement is most plausibly explained by dose-related behavioural suppression, likely reflecting physiological hypoactivity rather than conditioned aversion. Additionally, post-conditioning administration of 10 μ M ketamine produced a non-significant trend toward reducing methamphetamine-induced CPP, suggesting a possible dose-dependent modulatory effect that did not reach statistical significance in this invertebrate system. These findings provide the first systematic indication that ketamine's behavioural effects differ markedly from its reinforcing or anti-craving properties observed in mammals.

Across all groups in Experiment 1, ketamine exposure reduced preference for the drug-paired context, consistent with non-specific behavioural suppression. This pattern likely reflects hypoactivity rather than conditioned aversion. In mammals,

ketamine transiently alters locomotor activity by blocking NMDA receptors and disrupting calcium influx (Lisek et al., 2016; Zhang et al., 2021). In planaria, the same NMDA blockade may disrupt ventral cilia coordination, which is crucial for motility (King & Patel-King, 2016; Rompolas et al., 2013). Furthermore, ketamine has been shown to acutely induce more diffuse, non-cue-specific dopaminergic signalling (Wu et al., 2021). This generalised increase in dopamine may further diminish cue salience and blur the distinction between drug-paired and neutral contexts (Fitzpatrick & Morrow, 2017). Together, these mechanisms suggest the absence of CPP primarily reflects motor and dopaminergic suppression, warranting future studies combining CPP with quantitative movement tracking and neurochemical assays to clarify whether ketamine exerts genuine modulatory effects on reward encoding.

Building on these findings, methamphetamine at 10 μ M reliably elicited robust CPP in our sample, corroborating the species' capacity for forming strong cue-drug associations via dopaminergic pathways (Adams & Byrne, 2019; Hutchinson et al., 2015; Raffa et al., 2013). Post-conditioning administration of ketamine did not significantly reduce methamphetamine-induced CPP; however, the 10 μ M group showed a near-significant reduction ($p = .053$), with a large effect size ($d = 0.92$), suggesting partial, dose-dependent effects on disrupting memory consolidation. In contrast, the 30 μ M group showed minimal change, suggesting that higher concentrations predominantly produce general behavioural suppression rather than targeted modulation of reward coding.

However, the 30 μ M group showed a more minor CPP increase after methamphetamine conditioning ($M\ change = 48.89\ s$, $d = 1.21$) than the 10 μ M group ($M\ change = 203.08\ s$, $d = 1.94$). Both the 10 μ M and 30 μ M groups spent similar amounts of time in the methamphetamine-paired context during post-treatment testing

(See Table 2, Figure 5). These findings suggest that the non-significant effect in the 30 μ M group may reflect a combination of locomotor inhibition suppression and a reduced dynamic range of CPP, rather than an absence of ketamine modulation, supporting the interpretation of dose-dependent hypoactivity rather than selective disruption of conditioned preference.

Beyond these behavioural explanations, a key translational insight from this study is that the dissociative effects and anti-craving efficacy of ketamine diverge across species. In mammals, NMDA receptor blockade induces transient cortical disinhibition and glutamate surges that disrupt maladaptive salience attribution, enabling reorganisation of drug-cue memories through cortico-limbic-striatal integration (Ezquerro-Romano et al., 2018; Gerhard et al., 2020; Krystal et al., 2023). Planaria lack this network-level architecture, meaning NMDA antagonism may promote cellular plasticity without the cognitive decoupling needed to modify craving or learning.

Consequently, glutamatergic modulation manifests as physiological suppression rather than reconsolidation disruption. This distinction underscores that ketamine's anti-addictive efficacy likely involves more than NMDA blockade alone, potentially engaging vertebrate-specific mechanisms such as cortical network modulation and plasticity-promoting cascades. In invertebrate models like planaria, NMDA antagonism may primarily produce physiological suppression rather than the complex cognitive or anti-craving effects observed in mammals.

While neural complexity is a likely factor contributing to the lack of ketamine-induced CPP in planaria, several alternative explanations should also be considered. One possibility is that metabolic differences between planaria and vertebrates may result in altered pharmacokinetics, leading to rapid degradation or inactivation of

ketamine before it can exert behavioural effects (Sacavage et al., 2008). Additionally, drug permeability barriers, such as differences between the planaria epidermis and vertebrate blood-brain barriers, could limit the amount of ketamine that reaches neural targets, thereby reducing its efficacy in this model (Mekers et al., 2015).

Another important consideration is the potential divergence in NMDA receptor subunit composition or distribution in planaria (Ribeiro et al., 2005); if key subunits required for ketamine's actions are absent or functionally distinct, the drug may not produce the same rewarding effects observed in vertebrates. Taken together, these factors highlight the importance of considering species-specific physiological and molecular differences when interpreting negative results in cross-species pharmacological studies.

Literature Comparison

Ketamine's antidepressant and anti-addictive properties are well characterised in mammals; however, ketamine-induced CPP is inconsistent across studies. Some report CPP at higher doses (Suzuki et al., 1999; van der Kam et al., 2009), whereas others observe reinforcement only at low doses (Silva-Cardoso & Nobre, 2021; Liu et al., 2017). Sex- and age-dependent effects further complicate interpretation: female rats may show aversion to doses neutral in males (Strong et al., 2017), and adolescent males often fail to develop CPP altogether (Parise et al., 2013). The absence of CPP in planaria mirrors this cross-species variability and underscores the complexity of NMDA receptor pharmacodynamics. Additionally, unlike rodents, planaria are hermaphroditic and lack sexually dimorphic behaviours, meaning this model cannot capture sex-specific differences in drug response, which are critical for translating findings.

In contrast, methamphetamine reliably induces CPP in planaria, supporting conserved dopaminergic mechanisms across species (Amaning-Kwarteng et al., 2017; Hutchinson et al., 2015). Methamphetamine directly reverses dopamine transporters, increasing synaptic dopamine release and producing robust reinforcement (Yang et al., 2018). Ketamine's effects, involving multi-system glutamatergic-dopaminergic interactions, are therefore inherently less predictable in simplified neural architectures, possibly explaining our findings.

Limitations

The interpretive strength of this study is limited by the restricted range of ketamine concentrations tested (10 μ M and 30 μ M). This limitation may have masked important behavioural effects at lower or higher dosages, potentially complicating the differentiation between aversive, reinforcing, or non-specific motor-suppressing outcomes. Future studies should use a systematic, tiered dose-response protocol to define ketamine's behavioural effects in planaria better. This method will clarify the threshold for conditioned place preference and the onset of non-specific behavioural suppression. Should expanded testing continue to demonstrate neither reinforcement nor attenuation of preference across a broader dosing spectrum, this would provide additional validation for the hypothesis that NMDA antagonism alone does not induce reinforcing effects in invertebrate models. This outcome would indicate that the anti-addictive and antidepressant effects of ketamine rely on complex neural circuits that are not present in planaria.

Second, planaria were not re-exposed to methamphetamine following ketamine treatment, meaning any post-treatment reductions in preference could partially reflect spontaneous extinction. Prior studies indicate that CPP in planaria can naturally extinguish within 72 hours (Amaning-Kwarteng et al., 2017), suggesting that the

attenuation observed in Experiment 2 may not solely represent ketamine-mediated effects. Incorporating extinction-only controls, reinstatement, and delayed follow-up testing would help isolate pharmacological effects from procedural decay.

Third, inter-individual variability among wild-caught planaria likely contributed to behavioural heterogeneity. The sample consisted of individuals collected from multiple freshwater streams in the Wellington region of Aotearoa New Zealand, encompassing at least two unclassified phenotypes. Differences in environmental history, nutrition, or stress exposure may have influenced responsiveness to both methamphetamine and ketamine. Standardising genetic and environmental confounds – such as through clonal laboratory strains – would improve reproducibility and interpretation.

Fourth, behaviour was scored manually due to software constraints, which introduced potential observer bias and reduced temporal accuracy. This may have masked subtle shifts in locomotion or surface preference, and reductions in preference observed at both ketamine doses in Experiment 2 could partly reflect general hypoactivity rather than altered reward valuation. Automated motion-tracking systems capable of quantifying micro-movement and velocity would enhance detection sensitivity and permit elucidating reward learning from motor suppression.

Finally, the modest sample size ($n = 12$) limited statistical power. After ketamine treatment, the 10 μM group showed a reduction in methamphetamine-induced CPP with a large effect size ($d = 0.92$), although this change did not reach conventional statistical significance ($p = .053$). In contrast, the 30 μM group showed no meaningful change. This pattern indicates that the apparent trend in the 10 μM group may reflect a true behavioural phenomenon, but the modest sample size ($n = 12$ per group) limited

statistical power, increasing the risk of a Type II error. Consequently, while definitive conclusions cannot be drawn, these results suggest that low-dose ketamine could partially attenuate methamphetamine-induced CPP. Further investigation with a larger sample size is warranted to clarify its dose-dependent effects and translational potential.

Implications

Collectively, these findings delineate a translational boundary between invertebrate and mammalian models, with specific implications for the study of ketamine's pharmacology. The weak, non-significant attenuation of methamphetamine-induced CPP at 10 μ M ketamine provides tentative evidence that NMDA receptor antagonism can influence reward-context associations at a fundamental neural level. However, the lack of strong, dose-dependent effects and the presence of general behavioural suppression suggest that planaria lack the crucial neural structures, such as corticolimbic regions, necessary for ketamine's full anti-addictive impact seen in mammals. (Duman et al., 2019).

This trend highlights that while planaria offer a reliable platform for investigating basic dopaminergic reinforcement, their capacity to model the higher-order integrative and modulatory processes necessary for evaluating candidate glutamatergic treatments is restricted. Therefore, this observation demonstrates both a pharmacological effect and the inherent limitations of employing invertebrate models to predict mammalian responses to NMDA antagonists. Careful consideration of the cognitive, motivational, and motoric contributions to planaria behaviour is essential for interpreting preclinical findings and guiding the development of new therapies for substance use disorders.

Future Directions

Future investigations should move beyond behavioural metrics to examine the molecular and cellular correlates of NMDA receptor blockade in planaria. Although ketamine did not induce CPP, it may still have triggered intracellular cascades associated with synaptic plasticity, including upregulation of BDNF, activation of mTOR signalling, and changes in neoblast proliferation. Quantifying these markers through molecular assays could clarify whether the observed behavioural neutrality reflects limited circuit complexity or the absence of pharmacological action.

Parallel testing of other psychoactive compounds, such as psilocybin, may further delineate the conserved elements of neuroplastic modulation in simpler systems. Psilocybin, which acts as a 5-HT_{2A} partial agonist and promotes structural plasticity, could offer a serotonergic counterpart to the glutamatergic mechanisms investigated here (Agnorelli et al., 2025). Applying such compounds within CPP paradigms would yield valuable comparative insight into how different neurotransmitter systems modulate reward learning at varying levels of neural complexity.

Conclusion

Studies using simple organisms, such as planaria, provide a vital foundation for addiction research. By exploring the fundamental biological mechanisms that underlie drug reward, dependence, and intervention, these models help identify new molecular targets and preclinical screening strategies for potential therapies. These early-stage discoveries serve as a stepping stone toward developing more effective and innovative treatments in humans. Ultimately, sustained investment in basic research not only broadens our scientific understanding – it drives the translational breakthroughs needed to combat the ongoing public health challenges posed by substance use disorders.

This study represents the first systematic evaluation of ketamine's behavioural effects within the planaria CPP model. The findings demonstrate that ketamine, when administered in isolation, does not confer measurable reinforcing properties and may instead induce nonspecific behavioural suppression. Furthermore, administration of low-dose ketamine resulted in only a modest, non-significant dose-dependent reduction of methamphetamine-induced CPP. Collectively, these results establish planaria as a robust and sensitive model of dopaminergic reinforcement while delineating their substantial limitations for investigating cortical-glutamatergic mechanisms relevant to addiction. By clarifying the boundaries of translational validity in planaria, this research highlights critical considerations for extrapolating invertebrate findings to mammalian systems. These insights hold significant implications for the advancement of cross-species addiction models and underscore the necessity of higher-order neural architectures for evaluating candidate glutamatergic pharmacotherapies targeting substance use disorders.

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Appendix

Ketamine and Methamphetamine Solutions in Experiments 1 and 2

Table A1

Quantities for Solutions Used in Experiments 1 and 2

Solution	Stock Concentration (μM)	Volume of Stock (μL)	Volume of Planaria Water (mL)
Ketamine 10 μM (Experiment 1)	500	120	5.88
Ketamine 30 μM (Experiment 1)	500	360	5.64
Methamphetamine 10 μM (Experiment 2)	250	200	4.80
Ketamine 10 μM (Experiment 2)	500	60	2.94
Ketamine 30 μM (Experiment 2)	500	180	2.82